

Lab 11: Genomics & Biotechnology

BIOL-1

Name: _____ Date: _____

Overview

Biotechnology puts our knowledge of DNA to work. Today, scientists can **cut** DNA at precise locations, **copy** a single molecule into billions, **sort** fragments by size, **identify** individuals by their unique DNA fingerprint, and even **edit** genes to treat diseases. These tools have transformed medicine, forensics, agriculture, and our understanding of life itself.

In this lab, you will explore the core techniques of biotechnology — restriction enzymes, PCR, gel electrophoresis, DNA fingerprinting, genetic engineering, and CRISPR gene editing.

Learning Objectives

By the end of this lab, you will be able to:

1. Explain how restriction enzymes and ligase create recombinant DNA.
2. Describe the three steps of a PCR cycle and calculate DNA amplification.
3. Interpret a gel electrophoresis result.
4. Analyze a DNA fingerprint for a paternity test.
5. Describe the goals and ethical considerations of gene therapy and CRISPR.

Materials

- This worksheet
- Ruler (optional, for gel diagram questions)

Part 1: Cutting and Pasting DNA

In nature, restriction enzymes are bacterial defense weapons — they chop up invading viral DNA. Scientists figured out how to use them as precision molecular scissors to cut any DNA at specific sequences. DNA ligase acts as molecular glue to seal pieces back together.

1. What does a restriction enzyme do?

2. When a restriction enzyme cuts DNA, it often leaves short single-stranded overhangs called "sticky ends." Why are sticky ends useful for genetic engineering?

3. What does DNA ligase do?

4. What is recombinant DNA?

5. Put these steps in the correct order (number 1-5) to create a GMO bacterium that makes human insulin:

Step	Order (1-5)
Transform the recombinant plasmid into bacteria	
Cut the plasmid AND the human insulin gene with the same restriction enzyme	
Use DNA ligase to seal the human gene into the plasmid	
Select for bacteria that successfully took up the plasmid	
Mix the cut plasmid and cut human gene together so sticky ends match	

Part 2: PCR — The Molecular Photocopier

PCR (Polymerase Chain Reaction) can take a single molecule of DNA and make billions of copies in just a few hours. This is how forensic scientists can analyze a tiny drop of blood from a crime scene or how doctors can detect a virus in a patient's sample.

6. Fill in what happens at each step of one PCR cycle:

Step	Temperature	What Happens
Denaturation	~95°C	
Annealing	~55°C	
Extension	~72°C	

7. PCR uses Taq polymerase instead of normal human DNA polymerase. Why? (Hint: think about the denaturation temperature.)

8. DNA doubles with each PCR cycle. Fill in the table:

Cycle Number	Number of DNA Copies
Start	1
After cycle 1	
After cycle 2	
After cycle 3	
After cycle 5	
After cycle 10	

9. After 30 cycles, approximately how many copies would you have? (Use 2^{30})

Part 3: Gel Electrophoresis — Sorting DNA by Size

Gel electrophoresis uses an electric field to pull DNA through a gel matrix (like a molecular obstacle course). Small pieces slip through easily and travel far. Large pieces get stuck and stay near the top.

10. DNA has a negative charge (from its phosphate groups). In an electric field, does DNA migrate toward the positive (+) electrode or the negative (–) electrode?

11. Which fragments travel FARTHER through the gel — small ones or large ones?

12. You run four DNA samples on a gel. The results show:

- **Sample 1:** One band very close to the wells (top)
- **Sample 2:** One band near the bottom of the gel
- **Sample 3:** Three bands spread across the gel
- **Sample 4:** Matches the pattern of Sample 3 exactly

Questions:

a) Which sample contains the LARGEST DNA fragment(s)?

b) Which sample contains the SMALLEST DNA fragment(s)?

c) Samples 3 and 4 have the same banding pattern. What does this tell you about their DNA?

Part 4: DNA Fingerprinting

No two people (except identical twins) have exactly the same pattern of STRs (Short Tandem Repeats) — short DNA sequences repeated different numbers of times. This makes STR profiling the gold standard for forensic identification.

13. What are STRs? Why are they useful for identifying individuals?

14. Paternity test scenario: A child has STR bands at positions A, B, C, and D. The mother's bands are at positions A and C.

a) Which bands in the child MUST have come from the biological father?

b) Suspect 1 has bands at B, D, E, and F. Suspect 2 has bands at B, D, G, and H. Which suspect is more likely to be the biological father? Explain.

15. A crime scene DNA sample produces bands at positions 2, 5, and 8. Three suspects are tested. Only Suspect C has bands at positions 2, 5, and 8. What can you conclude?

Part 5: Gene Therapy and CRISPR

Gene therapy aims to treat genetic diseases by giving patients a working copy of a faulty gene. CRISPR-Cas9 takes this further — it can precisely find and edit specific DNA sequences, like "find and replace" for the genome.

16. What is gene therapy?

17. Compare ex vivo and in vivo gene therapy:

	Ex Vivo	In Vivo
Where are cells modified?		
Are cells removed from the patient?		

18. What is CRISPR-Cas9? How does it differ from simply using restriction enzymes?

19. What is the difference between somatic gene therapy and germline modification? Why is germline modification considered more ethically controversial?

20. The genetic code is nearly universal — the same codons code for the same amino acids in bacteria, plants, animals, and humans. Explain why this

universality makes biotechnology (like producing human insulin in bacteria) possible.

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